



Machine Learning

Dr. Shylaja S S

Professor

Department of Computer Science & Engineering

Acknowledgement : Dr. Rajanikanth K (Academic Advisor, PESU)
Prof. Preet Kanwal (Associate Professor, PESU)
Teaching Assistant (Arya Rajiv Chaloli)

After the initial feature engineering and implementing a model, we get some output in terms of a class or probability.

Effectiveness is measured by some metrics using our model and test data sets. Some **metrics for classification algorithms** are:

- Accuracy
- Precision
- Recall
- Specificity
- Receiver Operating Characteristics (ROC)
- Area Under Curve (AUC)

- Given a classifier and an instance, there are four possible outcomes.
 - **TP** - True Positive
If the instance is **positive** and is classified as **positive**
 - **FN** - False Negative
If the instance is **positive** and is classified as **negative**
 - **TN** - True Negative
If the instance is **negative** and is classified as **negative**
 - **FP** - False Positive
If the instance is **negative** and is classified as **positive**
- Confusion matrix (also called a contingency table) is a specific table layout that allows visualization of the performance of a classifier



Actual Label	Predicted Label	Result
+	+	True Positive
-	-	True Negative
+	-	False Negative
-	+	False Positive

		PREDICTIVE VALUES	
		POSITIVE (1)	NEGATIVE (0)
ACTUAL VALUES	POSITIVE (1)	TP	FN
	NEGATIVE (0)	FP	TN

The word **Positive/Negative** is used to indicate the **prediction made** by the model.

The word **True/False** is used to indicate whether the **prediction is right or wrong**.

Let's say we have data sets of 200 objects out of which 100 are from class A and 100 from class B

With predictions from our model as below

- 60 of the data object were correctly predicted positive class A as in the data set
- 30 of the data object were correctly predicted as negative class B as in the dataset

What is the total number of correct and wrong predictions?

N	class A	class B
200	100	100

N	class A	class B
200	100	100

- 60 of the data object were correctly predicted positive class A as in the dataset
- 30 of the data object were correctly predicted as negative class B as in the dataset

Total correct predictions: $60 + 30 = \mathbf{90}$
Total wrong predictions: $70 + 40 = \mathbf{110}$

N = 200

PREDICTIVE VALUES

POSITIVE (1) NEGATIVE (0)

ACTUAL VALUES

POSITIVE (1)

NEGATIVE (0)

TP 60	FN 40
FP 70	TN 30

- Accuracy in classification problems is the number of correct predictions made by the model over all kinds predictions made.
- Accuracy is generally a good measure when the target variable classes are nearly balanced.

- Accuracy =
$$\frac{TP + TN}{(TP + TN + FP + FN)}$$

		PREDICTIVE VALUES	
		POSITIVE (1)	NEGATIVE (0)
ACTUAL VALUES	POSITIVE (1)	TP	FN
	NEGATIVE (0)	FP	TN

	Predicted: (1)	Predicted: (0)
Actual: (1)	98	0
Actual: (0)	02	0

Accuracy is not a very good measure of the classifier when dataset is imbalanced.

Accuracy is generally a good measure when the **target variable classes are nearly balanced**.

Example:

Fraud Detection: Training data distribution

+ve - normal transaction (990)

-ve - fraudulent transaction (10)

In test data of 100 transactions: (if its a naive program where it calls every transaction as +ve), we achieve an accuracy of 98%

However, our interest here is to catch all the fraudulent transactions correctly.

- Precision: How many correct +ve cases did we catch?

- Precision =
$$\frac{TP}{(TP + FP)}$$

↑
In reality, the instance belonged to a negative class but was falsely predicted as +ve.

		PREDICTIVE VALUES	
		POSITIVE (1)	NEGATIVE (0)
ACTUAL VALUES	POSITIVE (1)	TP	FN
	NEGATIVE (0)	FP	TN

Precision is useful when the **cost of false positives** is significantly **higher** than the cost of false negatives

Examples:

- Spam detection:
Misclassifying a legitimate email as spam (false positive) can be highly inconvenient for the user, whereas missing a spam email (false negative) is less critical.
- Product recommendations:
Recommending items a user is unlikely to be interested in (false positive) can lead to user disengagement, while missing a few relevant recommendations (false negative) is less impactful.

- Recall: Sensitivity / True Positive Rate / Hit rate
- How many of the positive cases did we catch correctly?

- Recall =
$$\frac{TP}{(TP + FN)}$$

		PREDICTIVE VALUES	
		POSITIVE (1)	NEGATIVE (0)
ACTUAL VALUES	POSITIVE (1)	TP	FN
	NEGATIVE (0)	FP	TN

Recall is useful when the **cost of a false negative** is very high.

Examples:

- Medical diagnoses (especially for life-threatening conditions):
It's crucial to catch all potential cases, even if it leads to some false alarms. For example, a false negative in cancer detection could be devastating, while a false positive might only require further investigation.
- Fraud detection:
Banks may prefer to flag a few legitimate transactions as potentially fraudulent (false positives) than miss even a single fraudulent transaction.


$$\text{Arithmetic Mean} = \frac{x+y}{2}$$
$$\text{Harmonic Mean} = \frac{2xy}{x+y}$$

Let us find a measure that combines both recall and precision.

F1 Score: The harmonic mean of precision and recall.

F1 Score = Harmonic Mean(Precision, Recall)

The higher the F1 score the better, with 0 being the worst possible and 1 being the best.

It reaches its optimum 1 only if precision and recall are both at 100%.

Precision - Fraction of True Positives out of *Total Predicted Positives*

Recall - Fraction of True Positives out of *Total Actual Positives*

F1 is a balance between Precision and Recall. Model's capacity to predict positives correctly out of actual as well as predicted positives!

Any performance metric that uses values from both classes will be inherently sensitive to class skews.

Metrics such as accuracy, precision, and F score use values from both +ves and -ves of the confusion matrix.

- Specificity: True Negative Rate
- How many -ve cases did we catch correctly?

- Specificity =
$$\frac{TN}{TN + FP}$$

		PREDICTIVE VALUES	
		POSITIVE (1)	NEGATIVE (0)
ACTUAL VALUES	POSITIVE (1)	TP	FN
	NEGATIVE (0)	FP	TN

- TPR - True positive rate / Recall / Sensitivity = $TP / (TP + FN)$
of +ves correctly classified
- FNR - False negative rate = $FN / (TP + FN)$
of +ves incorrectly classified
- TNR - True negative rate / Specificity = $TN / (TN + FP)$
of -ves correctly classified
- FPR - False positive rate / 1 - Specificity = $FP / (TN + FP)$
of -ves incorrectly classified
- Precision - Fraction of True Positives out of Total Predicted Positives = $TP / (TP + FP)$
- Accuracy - # correct predictions made over all kind of predictions
- F1-Score - combined metric = $\text{harmonic_mean}(\text{recall}, \text{precision})$

Machine Learning

Confusion Matrix



		Predicted Class		
		Positive	Negative	
Actual Class	Positive	True Positive (TP)	False Negative (FN) Type II Error	Sensitivity $\frac{TP}{(TP + FN)}$
	Negative	False Positive (FP) Type I Error	True Negative (TN)	Specificity $\frac{TN}{(TN + FP)}$
		Precision $\frac{TP}{(TP + FP)}$	Negative Predictive Value $\frac{TN}{(TN + FN)}$	Accuracy $\frac{TP + TN}{(TP + TN + FP + FN)}$

Consider the following confusion matrix for a **binary classifier**
Let us calculate all the metrics we discussed till now

		Predicted Class	
		Positive	Negative
Actual Class	Positive	True Positive (TP) 100	False Negative (FN) Type II Error 5
	Negative	False Positive (FP) Type I Error 10	True Negative (TN) 50

Accuracy - Overall, how often is the classifier correct?

$$(TP+TN)/total = (100+50)/165 = 0.91$$

Precision: When it predicts yes, how often is it correct?

$$TP/predicted\ yes = 100/(100 + 10) = 0.91$$

Recall - When it's actually yes, how often does it predict yes?

$$TP/actual\ yes = 100/(100 + 5) = 0.95$$

Specificity - When it's actually no, how often does it predict no?

$$TN/actual\ no = 50/(10+50) = 0.83$$

$$\begin{aligned} \text{F1 Score} &= 2 * (\text{recall} * \text{precision}) / (\text{recall} + \text{precision}) \\ &= 2 * (0.95 * 0.91) / (0.95 + 0.91) = 0.93 \end{aligned}$$

		Predicted Class	
		Positive	Negative
Actual Class	Positive	True Positive (TP) 100	False Negative (FN) Type II Error 5
	Negative	False Positive (FP) Type I Error 10	True Negative (TN) 50

- As the name implies, it is a confusion matrix that deals with **multiple classes** (i.e. more than 2 classes).
- Just like the 2-class confusion matrix, it describes the performance of a multi-class classification model.

		Predicted		
		A	B	C
True labels	A	2	2	0
	B	1	2	0
	C	0	0	3

Actual	Elephant	25	3	0	2
	Monkey	3	53	2	3
	Fish	2	1	24	2
	Lion	1	0	2	71
		Elephant	Monkey	Fish	Lion
		Predicted			

Example - Multi-class confusion matrix

Let's assume that our multi-class classification model is one that classifies images of dogs into the following breeds:

Husky

Labrador

Bulldog



Notation Used :
P(Actual, Prediction)

		Predicted		
		Husky	Labrador	Bulldog
Actual	Husky	P(H,H)	P(H,L)	P(H,B)
	Labrador	P(L,H)	P(L,L)	P(L,B)
	Bulldog	P(B,H)	P(B,L)	P(B,B)

Metrics calculations - Multi-class confusion matrix

For Husky class

True Positive	$P(H,H)$
True Negative	$P(L,L) + P(L,B) + P(B,L) + P(B,B)$
False Positive	$P(L,H) + P(B,H)$
False Negative	$P(H,L) + P(H,B)$



Notation Used :
 $P(\text{Actual, Prediction})$

		Predicted		
		Husky	Labrador	Bulldog
Actual	Husky	$P(H,H)$	$P(H,L)$	$P(H,B)$
	Labrador	$P(L,H)$	$P(L,L)$	$P(L,B)$
	Bulldog	$P(B,H)$	$P(B,L)$	$P(B,B)$

Metrics calculations - Multi-class confusion matrix

For Labrador class

True Positive	$P(L,L)$
True Negative	$P(H,H) + P(H,B) + P(B,H) + P(B,B)$
False Positive	$P(H,L) + P(B,L)$
False Negative	$P(L,H) + P(L,B)$



Notation Used :
 $P(\text{Actual, Prediction})$

		Predicted		
		Husky	Labrador	Bulldog
Actual	Husky	$P(H,H)$	$P(H,L)$	$P(H,B)$
	Labrador	$P(L,H)$	$P(L,L)$	$P(L,B)$
	Bulldog	$P(B,H)$	$P(B,L)$	$P(B,B)$

Metrics calculations - Multi-class confusion matrix

For Bulldog class

True Positive	$P(B,B)$
True Negative	$P(H,H) + P(H,L) + P(L,H) + P(L,L)$
False Positive	$P(H,B) + P(L,B)$
False Negative	$P(B,H) + P(B,L)$



Notation Used :
 $P(\text{Actual}, \text{Prediction})$

		Predicted		
		Husky	Labrador	Bulldog
Actual	Husky	$P(H,H)$	$P(H,L)$	$P(H,B)$
	Labrador	$P(L,H)$	$P(L,L)$	$P(L,B)$
	Bulldog	$P(B,H)$	$P(B,L)$	$P(B,B)$

Accuracy calculation - multi-class confusion matrix

Accuracy =
$$\frac{P(H,H) + P(L,L) + P(B,B)}{(P(H,H) + P(L,H) + P(B,H) + P(H, L) + P(L,L) + P(B,L) + P(H, B) + P(L,B) + P(B,B))}$$

Let us calculate Accuracy of the Classifier

Notation Used :
P(Actual, Prediction)

		Predicted		
		Husky	Labrador	Bulldog
Actual	Husky	P(H,H)	P(H,L)	P(H,B)
	Labrador	P(L,H)	P(L,L)	P(L,B)
	Bulldog	P(B,H)	P(B,L)	P(B,B)

Precision Calculation - multi-class confusion matrix

For Husky class

True Positive	$P(H,H)$
True Negative	$P(L,L) + P(L,B) + P(B,L) + P(B,B)$
False Positive	$P(L,H) + P(B,H)$
False Negative	$P(H,L) + P(H,B)$

For Labrador class

True Positive	$P(L,L)$
True Negative	$P(H,H) + P(H,B) + P(B,H) + P(B,B)$
False Positive	$P(H,L) + P(B,L)$
False Negative	$P(L,H) + P(L,B)$

For Bulldog class

True Positive	$P(B,B)$
True Negative	$P(H,H) + P(H,L) + P(L,H) + P(L,L)$
False Positive	$P(H,B) + P(L,B)$
False Negative	$P(B,H) + P(B,L)$

Precision = $\frac{\text{True positives}}{\text{True positives} + \text{False Positives}}$

Precision (Husky) = $\frac{P(H,H)}{P(H,H)+P(L,H)+P(B,H)}$

Precision (Labrador) = $\frac{P(L,L)}{P(L,L)+P(H,L)+P(B,L)}$

Precision (Bulldog) = $\frac{P(B,B)}{P(B,B)+P(H,B)+P(L,B)}$

Recall/TPR Calculation - multi-class confusion matrix

For Husky class

True Positive	$P(H,H)$
True Negative	$P(L,L) + P(L,B) + P(B,L) + P(B,B)$
False Positive	$P(L,H) + P(B,H)$
False Negative	$P(H,L) + P(H,B)$

For Labrador class

True Positive	$P(L,L)$
True Negative	$P(H,H) + P(H,B) + P(B,H) + P(B,B)$
False Positive	$P(H,L) + P(B,L)$
False Negative	$P(L,H) + P(L,B)$

For Bulldog class

True Positive	$P(B,B)$
True Negative	$P(H,H) + P(H,L) + P(L,H) + P(L,L)$
False Positive	$P(H,B) + P(L,B)$
False Negative	$P(B,H) + P(B,L)$

Recall = $\frac{\text{True positives}}{\text{True positives} + \text{False Negatives}}$

Recall (Husky) = $\frac{P(H,H)}{P(H,H)+P(H,L)+P(H,B)}$

Recall (Labrador) = $\frac{P(L,L)}{P(L,L)+P(L,H)+P(L,B)}$

Recall (Bulldog) = $\frac{P(B,B)}{P(B,B)+P(B,H)+P(B,L)}$

Multi-class confusion matrix - Practice Problem

Now you know how to calculate the metric for n classes.

Evaluate the given confusion matrix...

		Predicted			
		A	B	C	D
Actual	A	100	80	10	10
	B	0	9	0	1
	C	0	1	8	1
	D	0	1	0	9

Multi-class confusion matrix - Practice Problem

Solution :
Accuracy = $\frac{100+9+8+9}{230}$

TP(A) = 100,
TP(B) = 9,
TP(C) = 8,
TP(D) = 9,

FP(A) = 0,
FP(B) = 82,
FP(C) = 10,
FP(D) = 12,

FN(A) = 100,
FN(B) = 1,
FN(C) = 2,
FN(D) = 1,

PRECISION(A) = 1,
PRECISION(B) = $\frac{9}{91}$,
PRECISION(C) = $\frac{8}{18}$,
PRECISION(D) = $\frac{9}{21}$,

Recall(A) = $\frac{1}{2}$
Recall(B) = $\frac{9}{10}$
Recall(C) = $\frac{8}{10}$
Recall(D) = $\frac{9}{10}$

Avg Precision = 0.492,

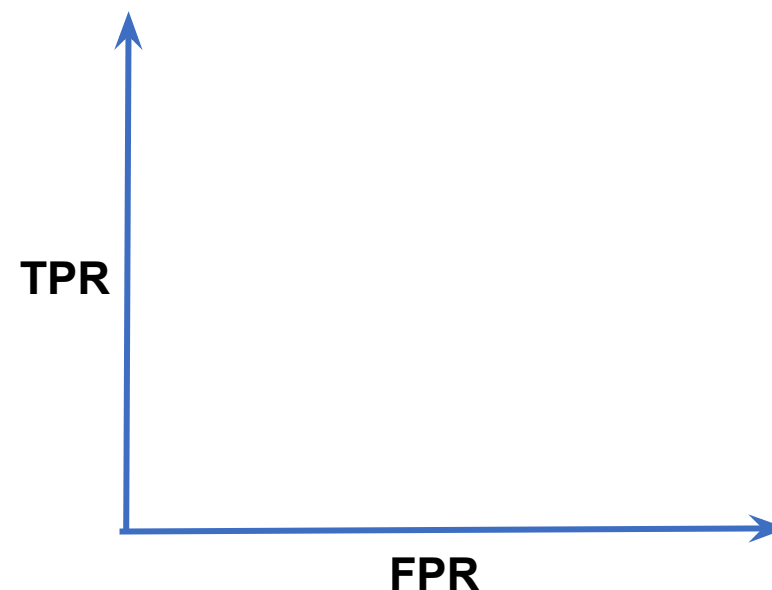
Avg Recall = 0.775,

F1-Score = 0.601

		Predicted			
		A	B	C	D
Actual	A	100	80	10	10
	B	0	9	0	1
	C	0	1	8	1
	D	0	1	0	9

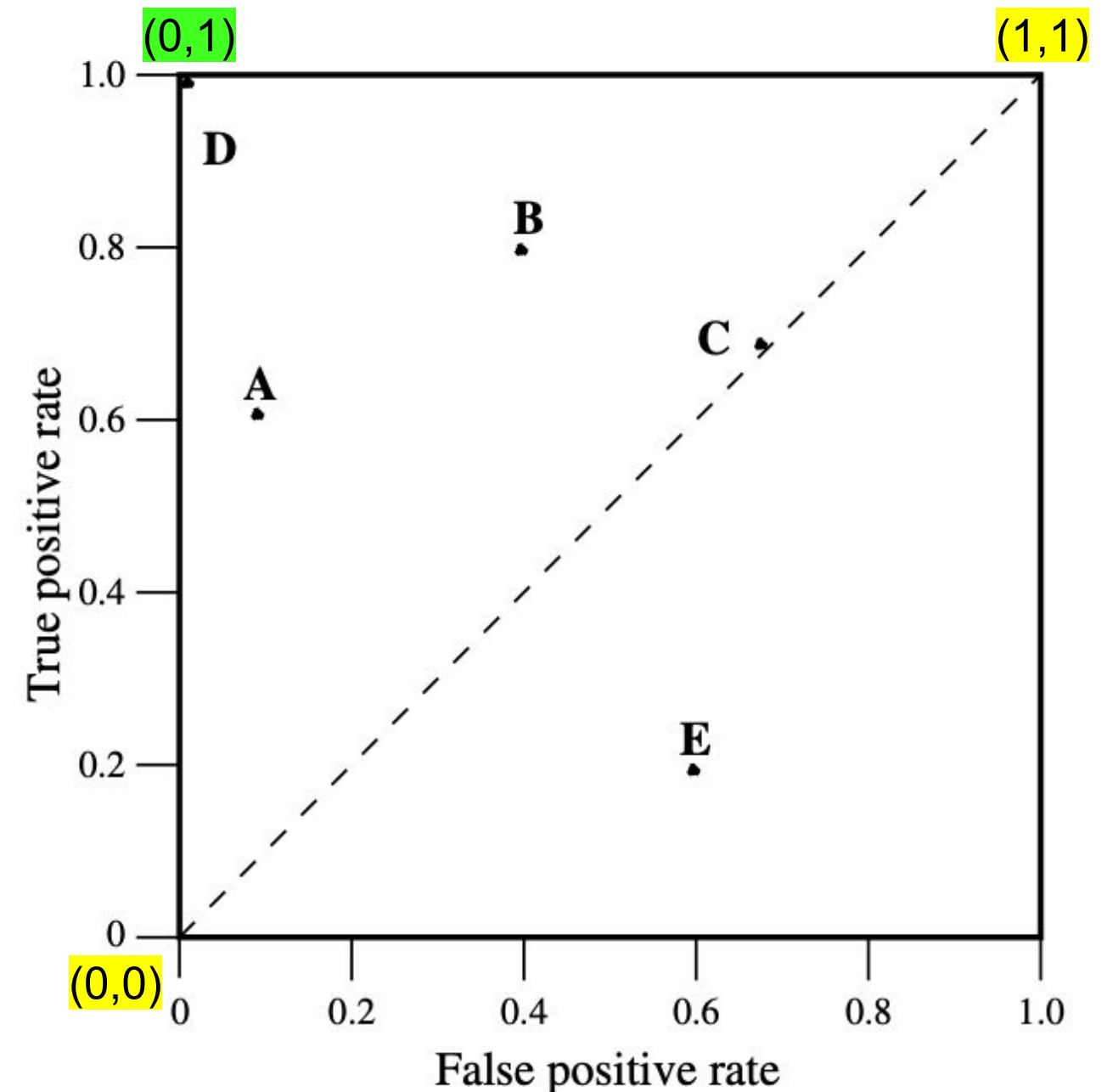
ROC - Receiver Operating Characteristics

- ROC is a probability curve that plots the TPR against FPR at various threshold values and essentially separates the 'signal' from the 'noise'.
- An ROC graph depicts relative tradeoffs between benefits (true positives) and costs (false positives).

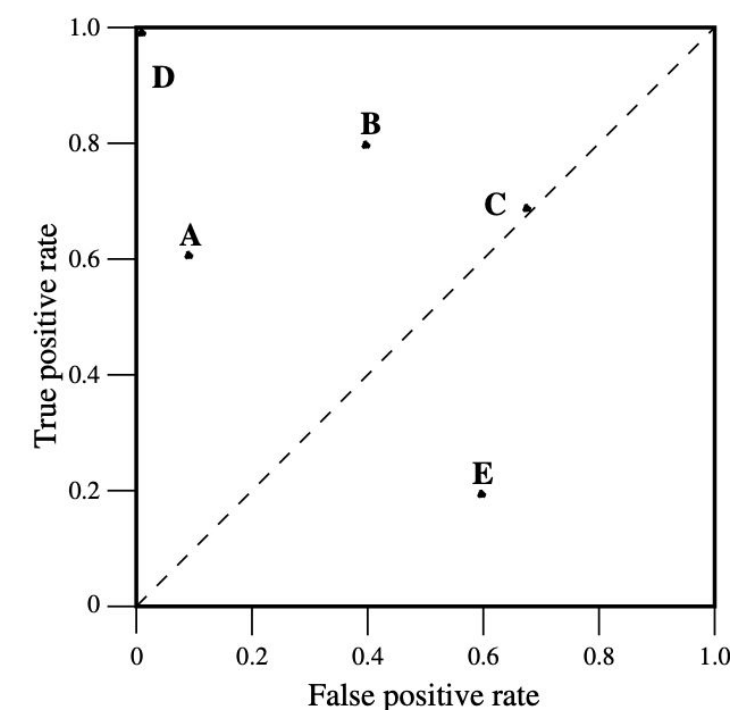
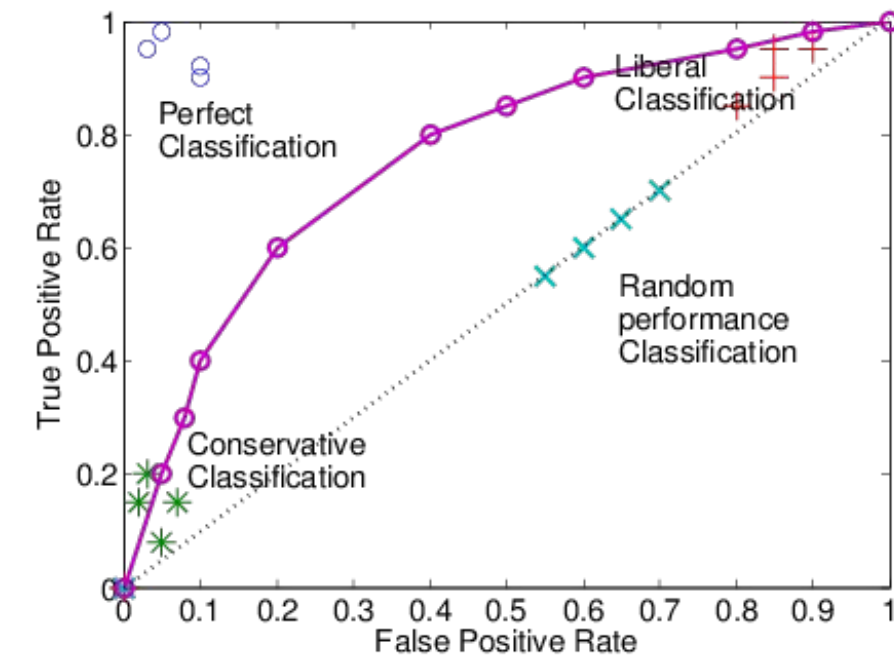


Points in ROC space are important to note.

- **(0, 0)** - never issuing a positive classification; such a classifier commits no false positive errors but also gains no true positives.
- **(1, 1)** - always issuing a positive classification
- **(0, 1)** - represents a perfect classification
- Diagonal line **$y=x$** represents the strategy of randomly guessing a class.
- Classifiers in the lower right triangle performs worse than random guessing.

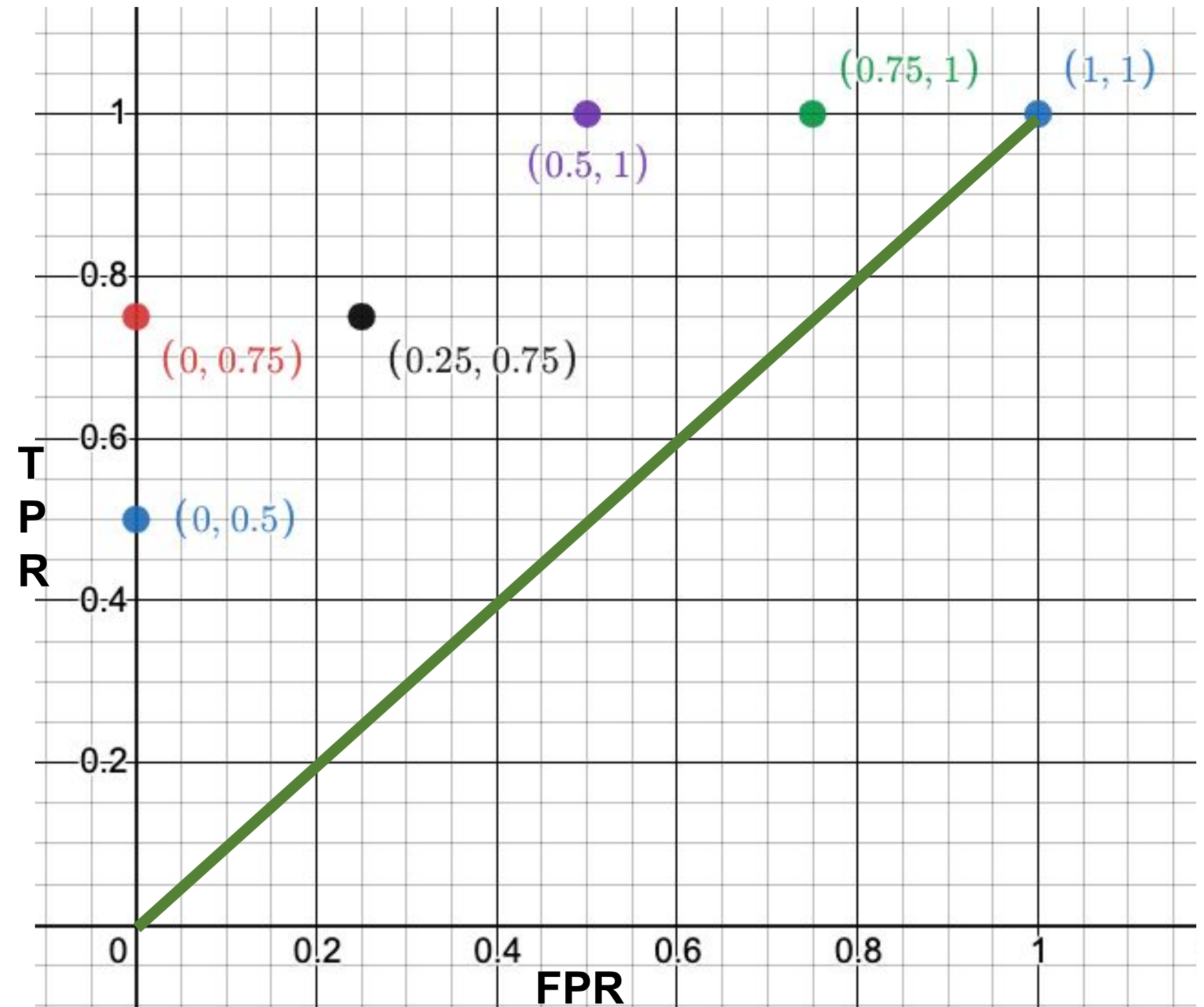


- A point in ROC space is better than another if it is to the northwest (tp rate is higher, fp rate is lower, or both – towards (0,1)) of the first.
- **Conservative classifiers** tend to have lower true positive rates and lower false positive rates, indicating a preference for strong evidence before making positive classifications.
- **Liberal classifiers** exhibit higher true positive rates and higher false positive rates, suggesting a tendency to classify more instances as positive, even with weaker evidence.
- Example: A is more conservative than B.



Take the example of an obesity classifier

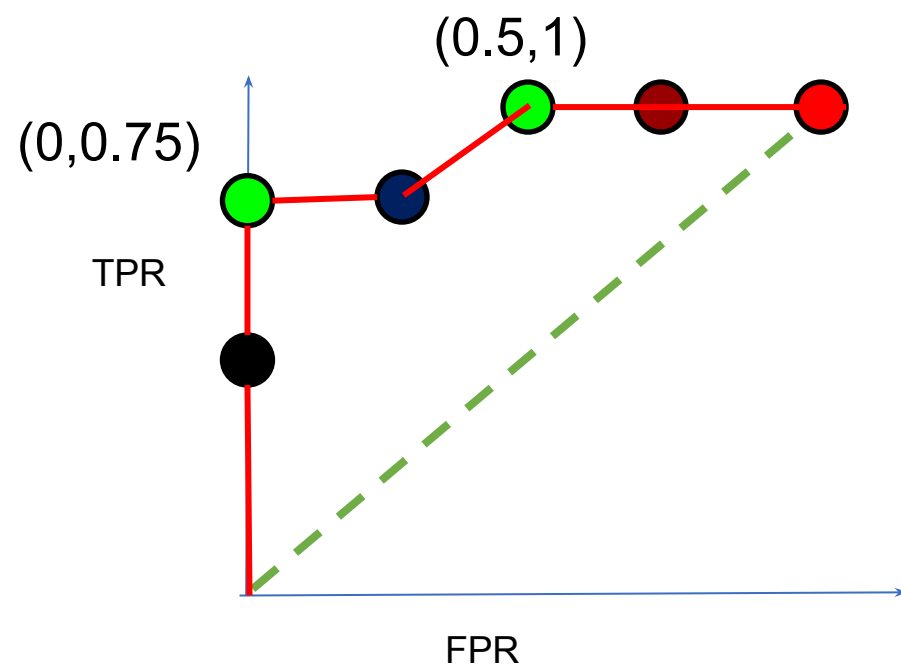
Threshold	FPR	TPR
0 (all sample classified obese)	1	1
0.3	0.75	1
0.4	0.5	1
0.6	0.25	0.75
0.7	0	0.75
0.9	0	0.5

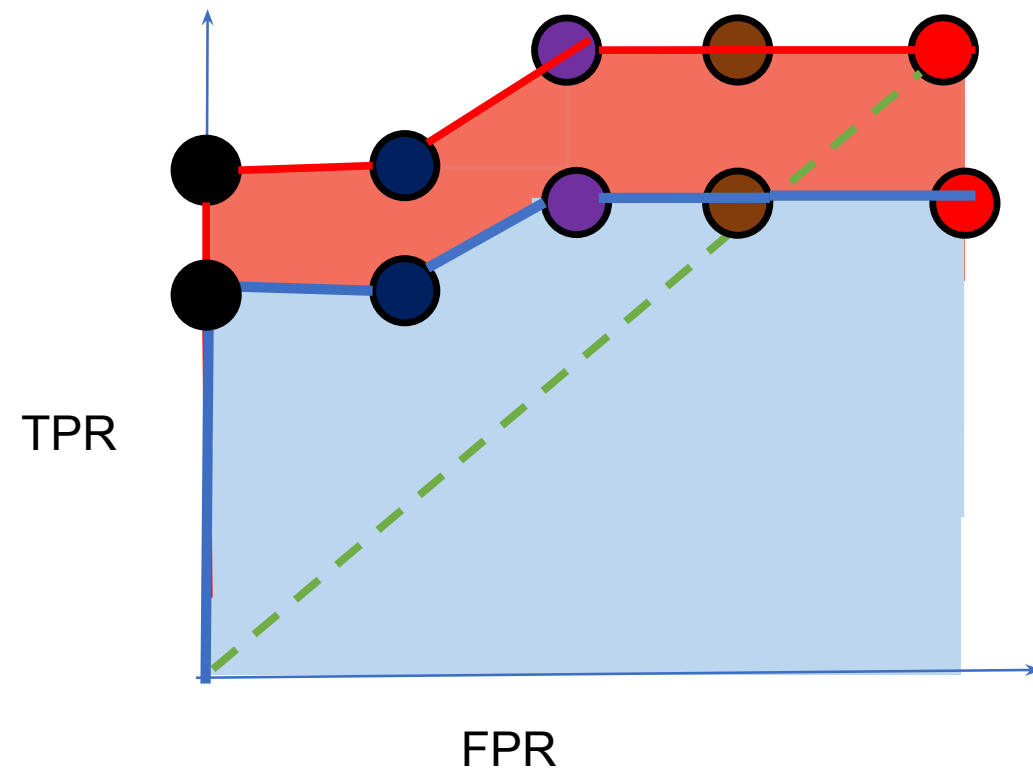


Threshold	FPR	TPR
0 (all sample classified obese)	1	1
0.3	0.75	1
0.4	0.5	1
0.6	0.25	0.75
0.7	0	0.75
0.9	0	0.5

The ROC graph summarizes all of the confusion matrix that each threshold produced

From the graph depending on how many false positive we are willing to accept the optimal one will be the point (0,0.75) or (0.5,1)





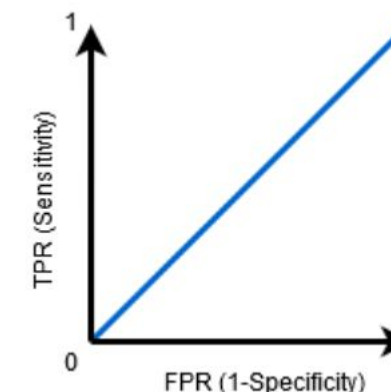
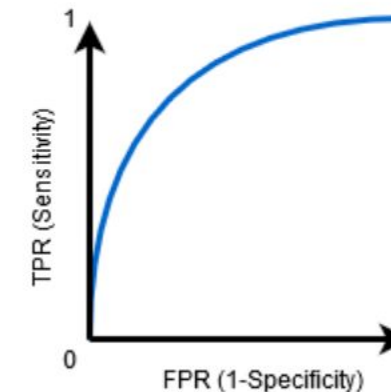
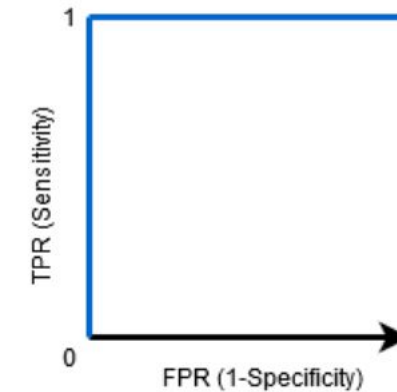
Now that we know what is ROC graph let us discuss the Area Under the Curve (AUC)

The AUC for the red ROC curve is greater than the AUC for the blue which suggests method 1 is better than method two

The AUC defines how good is our model, **more the AUC better is the model**. This helps us in comparing two models and selecting best among them for our problem

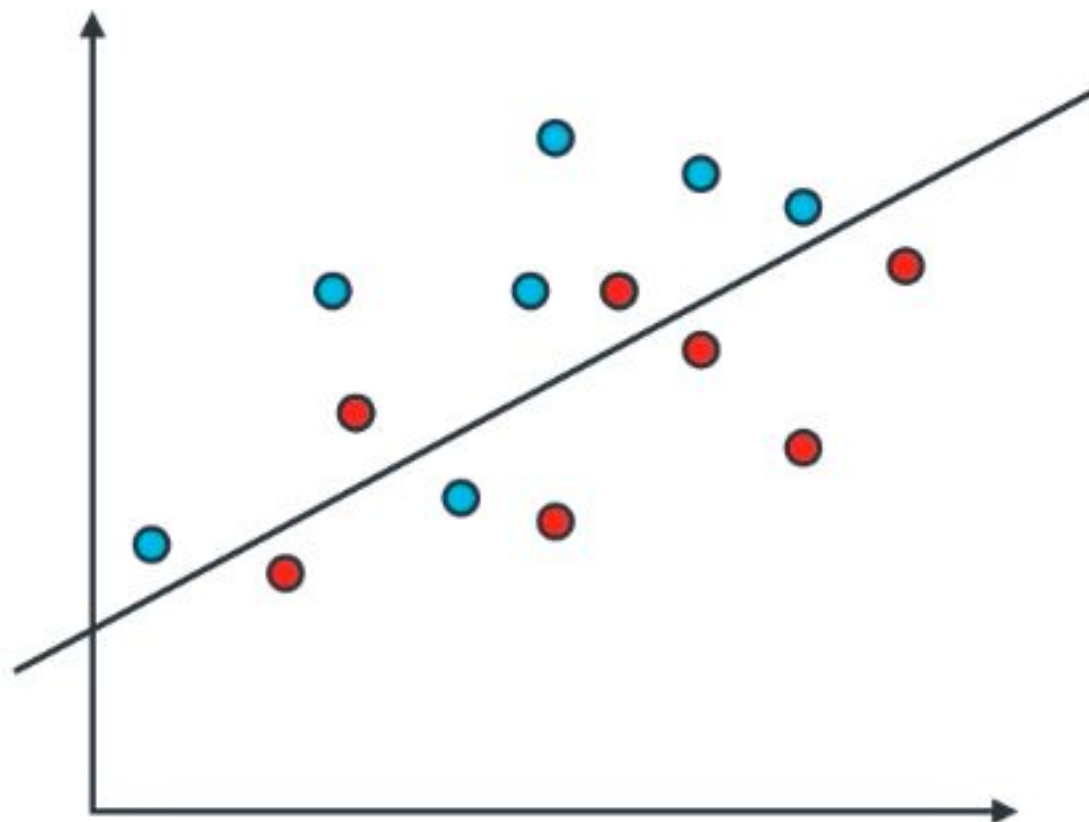
The higher the AUC, the better the performance of the model at distinguishing between the positive and negative classes.

- When **AUC = 1**, then the classifier is able to perfectly distinguish between all the Positive and the Negative class points correctly.
- If, however, the **AUC had been 0**, then the classifier would be predicting all Negatives as Positives, and all Positives as Negatives.
- When **$0.5 < \text{AUC} < 1$** , there is a high chance that the classifier will be able to distinguish the positive class values from the negative class values.
- When **AUC=0.5**, then the classifier is not able to distinguish between Positive and Negative class points. Meaning either the classifier is predicting random class or constant class for all the data points.



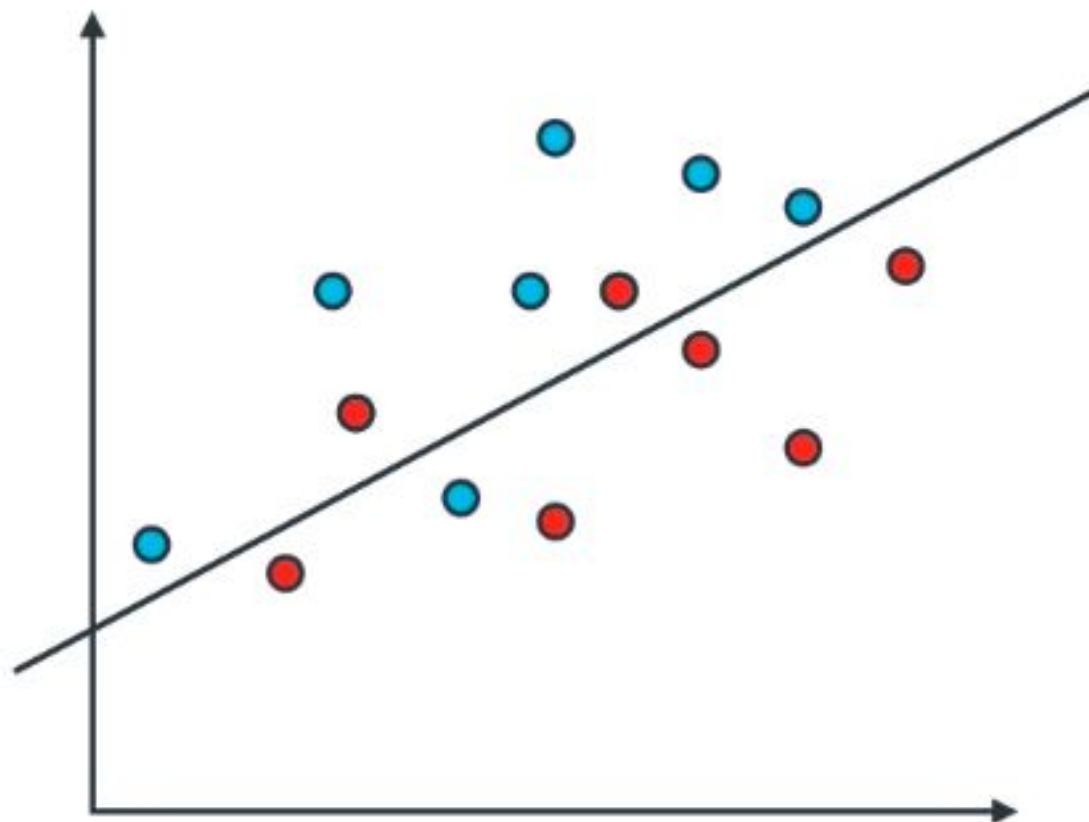
Practice Questions

Find the number of TP, FP, TN, FN cases for the given graph, where blue is positive and red is negative



		Prediction	
		Guessed Positive	Guessed Negative
Data	Positive		
	Negative		

Find the number of TP, FP, TN, FN cases for the given graph, where blue is positive and red is negative



		Prediction	
		Guessed Positive	Guessed Negative
Data	Positive	6 True positives	1 False Negatives
	Negative	2 False Positives	5 True Negatives



Accuracy

Patients	Diagnosis	
	Diagnosed sick	Diagnosed Healthy
	Sick	Healthy
Sick	1000	200
Healthy	800	8000

Accuracy: Out of the all the patients, how many did we classify correctly?

$$\text{Accuracy} = \frac{1,000 + 8,000}{1000 + 200 + 800 + 8000}$$

$$= 90\%$$



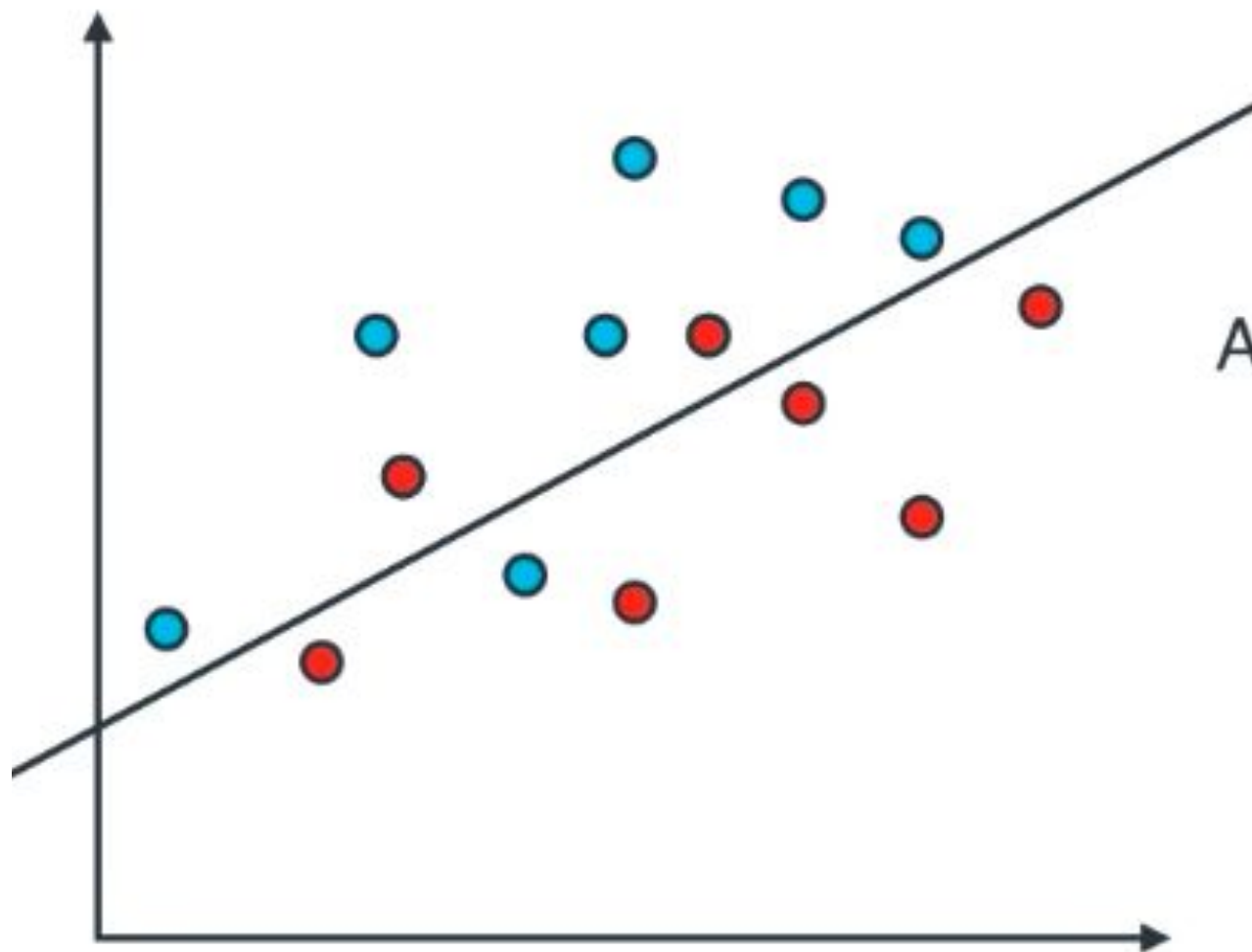
Accuracy

E-mail	Folder	
	Spam Folder	Inbox
Spam	100	170
Not spam	30	700

Accuracy: Out of the all the e-mails, how many did we classify correctly?

$$\text{Accuracy} = \frac{100 + 700}{100 + 170 + 30 + 700}$$

$$= 80\%$$



Precision: Out of all the data, how many points did we classify correctly?

$$\text{Accuracy} = \frac{\text{Correctly Classified points}}{\text{All points}}$$



Precision

Diagnosis

Patients	Diagnosis	
	Sick	Healthy
	Diagnosed sick	Diagnosed Healthy
Sick	1000	200 ❌
Healthy	800	8000

Precision: Out of the patients we diagnosed with an illness, how many did we classify correctly?

$$\text{Precision} = \frac{1,000}{1,000 + 800} = 55.7\%$$

Precision

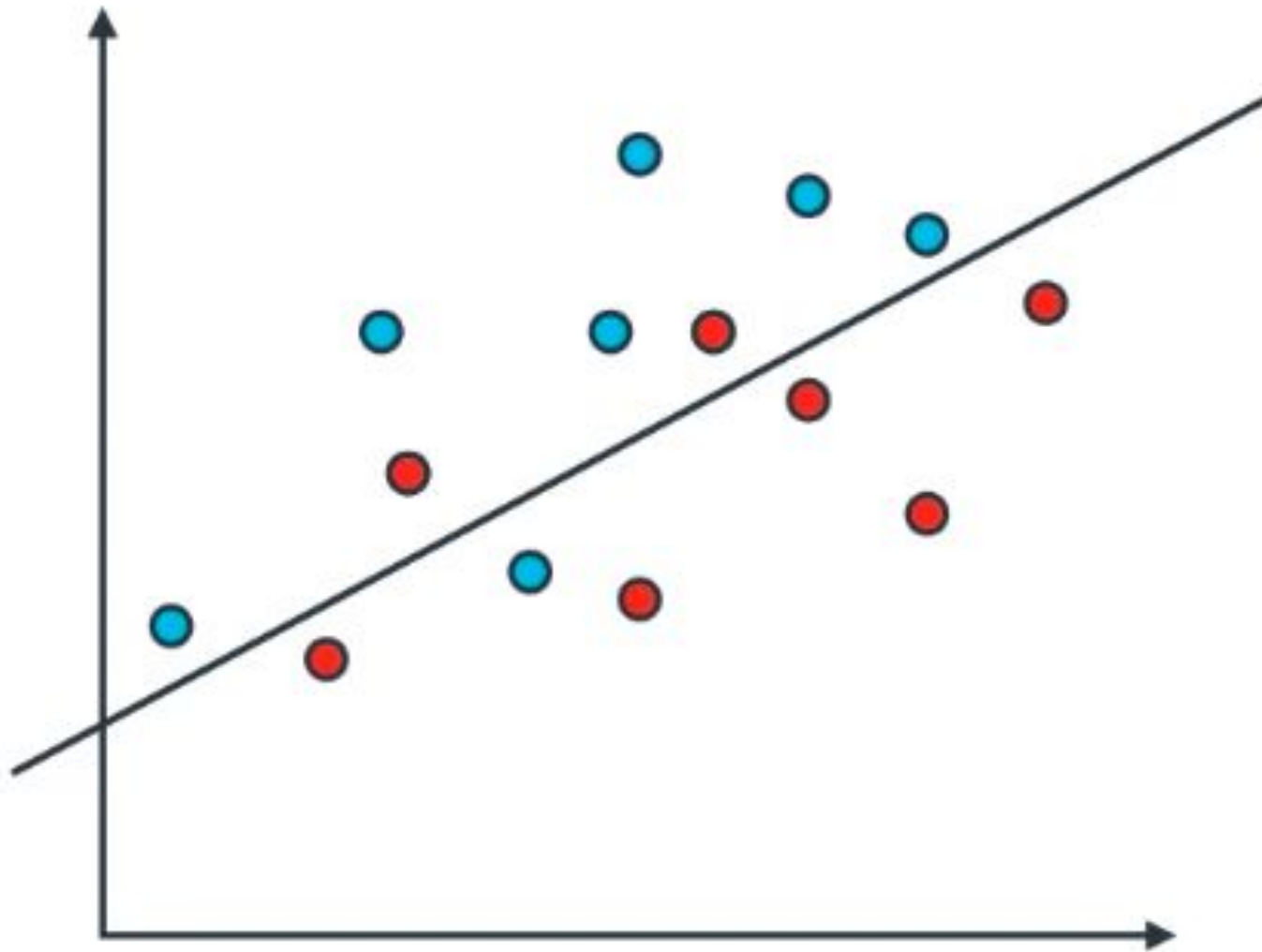


Folder

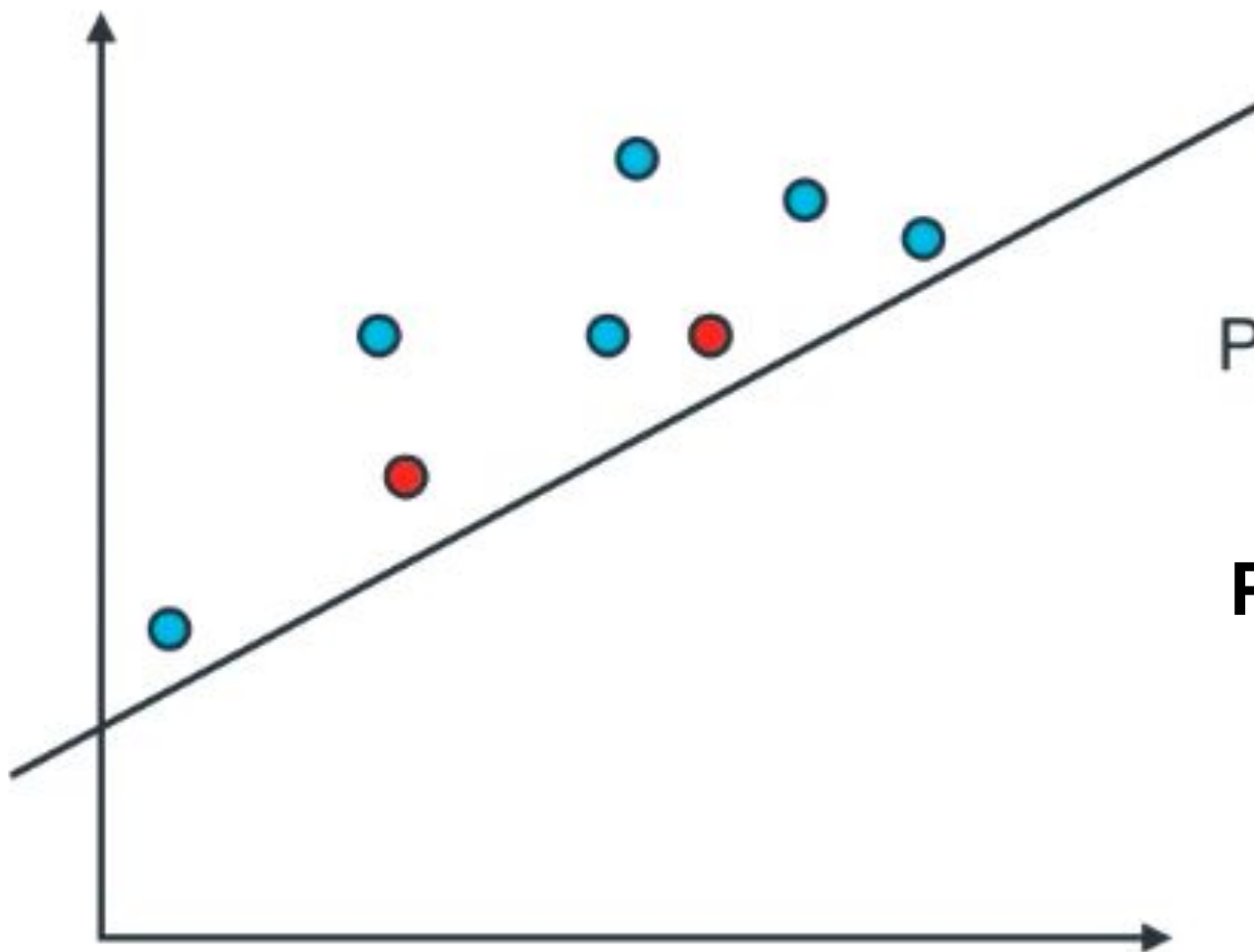
E-mail		Spam Folder	Inbox
	Spam	100	170
	Not spam	30 	700

Precision: Out of the all the e-mails, sent to the spam inbox, how many were actually spam?

$$\text{Precision} = \frac{100}{100 + 30} = 76.9\%$$



Precision: Out of the points we've predicted to be positive, how many are correct?



Precision: Out of the points we've predicted to be positive, how many are correct?

$$\text{Precision} = \frac{\text{True positives}}{\text{True positives} + \text{False Positives}}$$

$$\text{Precision} = 6/8$$

Machine Learning

Recall - Example

Recall



		Diagnosis	
		Diagnosed Sick	Diagnosed Healthy
Patients	Sick	1000	200 ✖
	Is Healthy	800	8000

Recall: Out of the sick patients, how many did we correctly diagnose as sick?

$$\text{Recall} = \frac{1,000}{1,000 + 200} = 83.3\%$$

Machine Learning

Recall - Example



Recall

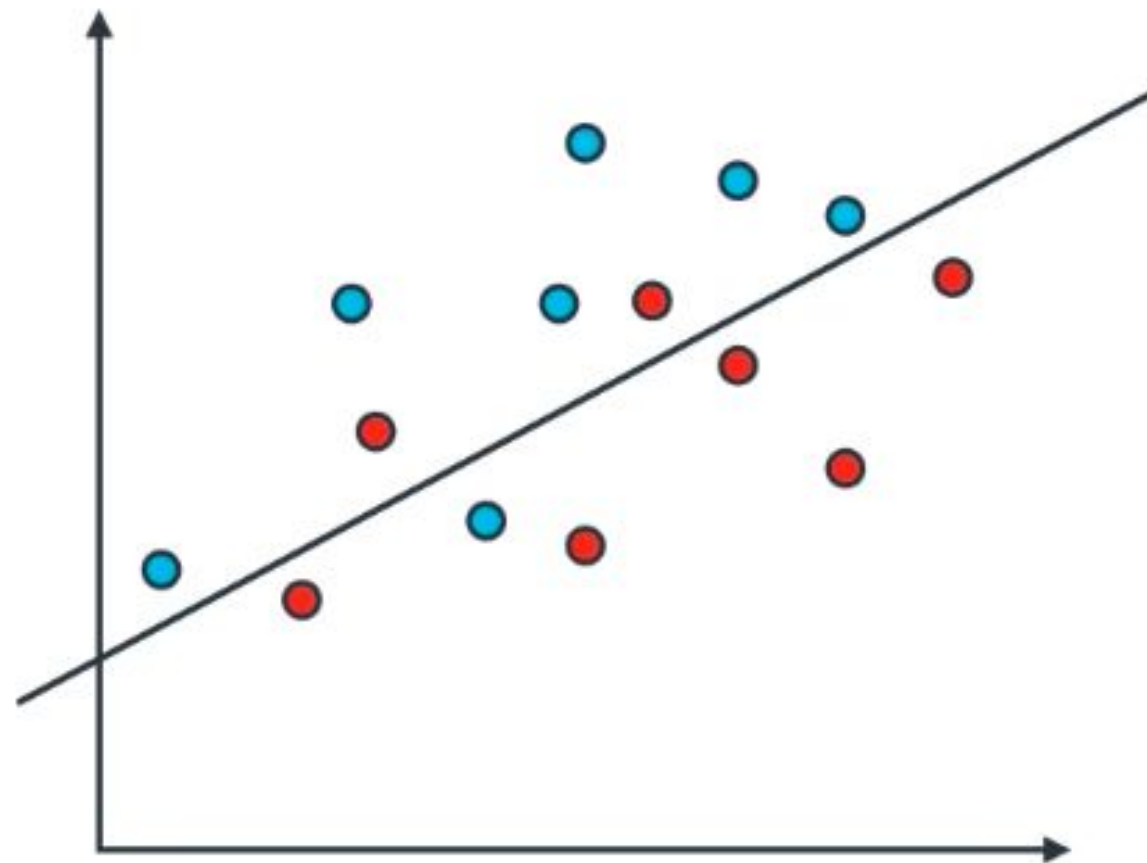
		Folder	
		Spam Folder	Inbox
E-mail	Spam	100	170
	Not spam	30 	700

Recall: Out of the all the spam e-mails, how many were correctly sent to the spam folder?

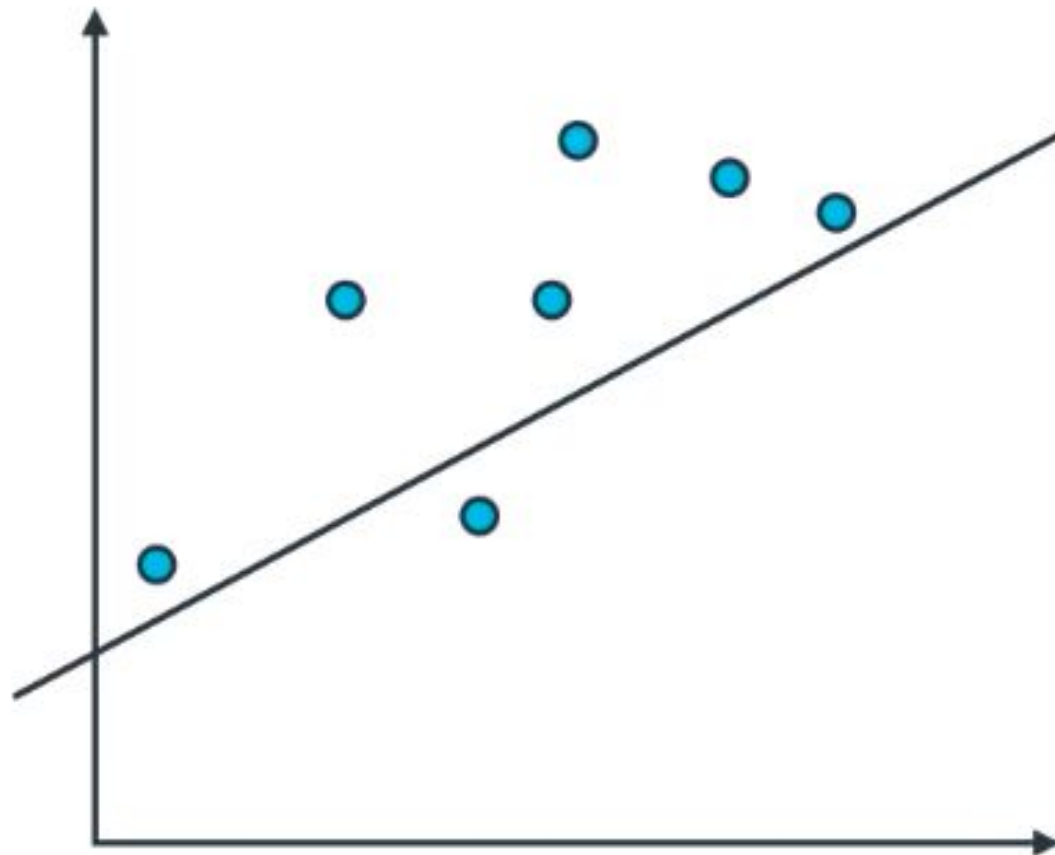
$$\text{Recall} = \frac{100}{100 + 170} = 37\%$$

Machine Learning

Recall - Question



Recall: Out of the points labelled positive, how many did we correctly predict?



Recall: Out of the points labelled positive, how many did we correctly predict?

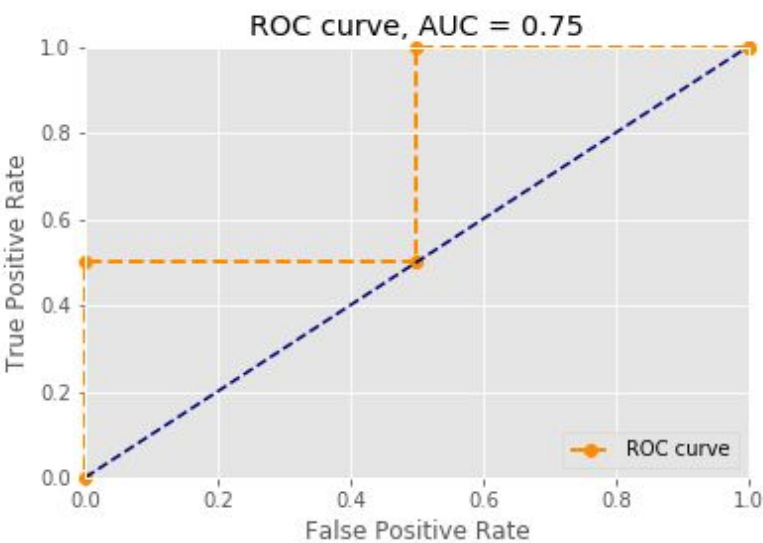
$$\text{Recall} = \frac{\text{True positives}}{\text{True positives} + \text{False Negatives}}$$

$$\text{Recall} = 6/7$$

Given the data, plot ROC Curve

Actual Class (y)	Predicted Probabilities ($\hat{y}$)	Threshold					
		$(\hat{y}_0)$	$(\hat{y}_{0.2})$	$(\hat{y}_{0.4})$	$(\hat{y}_{0.6})$	$(\hat{y}_{0.8})$	$(\hat{y}_1)$
1	0.8	1	1	1	1	1	0
0	0.6	1	1	1	1	0	0
1	0.4	1	1	1	0	0	0
0	0.2	1	1	0	0	0	0

FPR	1	1	0.5	0.5	0	0
TPR	1	1	1	0.5	0.5	0





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Dr. Shylaja S S

Department of Computer Science & Engineering

shylaja.sharath@pes.edu